ROHIT BANKAR

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EDUCATION

Purdue University *January 2024 - December 2025*
Master of Science in Computer and Information Technology ***(4.0/4.0)*** *West Lafayette, IN*

Indian Institute of Technology (IIT) Indore *July 2018 - May 2022*
Bachelor's in Technology (BTech) **(8.77/10.0)**

PROFESSIONAL EXPERIENCE

Amazon (AWS Bedrock) *February 2026 - Present*
Software Development Engineer *Seattle, WA*

- Engineer control-plane services in Forklift Forge for Anthropic Claude deployments on AWS Bedrock, managing replica lifecycle, deployment workflows, and distributed capacity orchestration.
- Worked on a metrics export pipeline using Amazon Kinesis, AWS Lambda, and CloudWatch, enabling secure cross-organization model telemetry sharing and replica-level alarm monitoring to improve observability and reliability.

Amazon (AWS Bedrock) *May 2025 - August 2025*
Software Development Intern *Seattle, WA*

- Developed **Hubble**, a centralized service for managing **Anthropic Claude LLM** models (Opus, Sonnet, Haiku) on AWS Bedrock, integrating with Trainium-based infrastructure to optimize deployments and reduce manual intervention by 90%.
- Enhanced the Forklift team's operational efficiency by optimizing **inference latency, reducing throttling, and integrating API-based capacity management with automated rebalance planning**, reducing issue-resolution time by 50%**.**
- Implemented and integrated **15+ RESTful APIs,** enhancing visibility and management efficiency for Claude replicas.

Razorpay (South Asia's Largest Payment Gateway) *July 2022 - December 2023*
Software Development Engineer (Backend) *Bengaluru, India*

- Interacted with 5+ teams to lead the re-architecture of Harvester, a real-time Spark-based data service, **saving $10-11k /month.**
- Coded custom whitelisting logic for Maxwell and Debezium CDC collectors, securing ~90% of company data.
- Migrated SRM dashboard **from TiDb to Pinot, reducing lag from 5 to 1 minute**, enhancing experience of 5+ teams & clients.
- Integrated Schema Registry into entity onboarding flows and ETL pipelines by implementing necessary transformation logic and handling horizontal and vertical scaling of pods, resulting in better versioning and 100% faster retrieval of schemas.
- Improved on-call bot's data flow, adding features for real-time updates, achieving 40% faster resolution.

Jaguar Land Rover *June 2021 - August 2021*
Software Development Intern *Bengaluru, India*

- Designed and coded a Machine Learning Pipeline to extract critical information from handwritten sentences using standard NLP libraries **like Stanford-openie and Textacy, achieving 70%** improvement over previous algorithms.
- Made exponential smoothing and **LSTM-RNN models, boosting failure prediction accuracy from 80% to over 90%.**
- Wrote a model to quantify optically scanned data using string-matching algorithms like Levenshtein-Damerau and Jaro-Winkler, achieving desired results with 95% accuracy.

RESEARCH & PROJECTS

Purdue University - Graduate Research Assistant *January 2024 - December 2025*
ML Systems and Infrastructure Research *West Lafayette, IN*

- Designed a predictive ML resource metamodel to estimate CPU, GPU, memory, and training time requirements, reducing infrastructure costs by 25% and enabling data-driven capacity planning.
- Proposed and implemented CORE-ML (resource allocation algorithm) and DeepScale (Kubernetes VPA-based autoscaling), reducing CPU over-provisioning by up to 80% & compute costs by up to 60%; published at IEEE ANTS 2025 & ICC 2026.
- Built and maintained 15+ production-grade data pipelines integrating Qualtrics and Salesforce with on-prem databases, processing thousands of daily records and delivering analytics dashboards to 20+ stakeholders.
- Developed and maintained a scalable web platform for the NSF-funded Advancing Cybersecurity Education initiative.

Detection of Suicidal Ideation through Deep Learning *March 2023 – October 2023*
Research Project - IIT Indore

- Developed a model using BERT, Graph Sage, and Hierarchical Attention to classify suicidal texts with **an F1 score > 0.975.**
- Enhanced model accuracy by integrating lexicons and emotional analysis, achieving a total F1 score increase of 6-7%.

Open-Source Contributions *January 2021 – December 2023*

- Programmed a Go authentication module for **Trino gateway**, enhancing security & access control for 2000+ Razorpay users.
- Integrated credential authentication into Query Book's UI, benefiting 10000+ users.
- Engineered a tool to convert LookML to Cube.js at Razorpay's annual hackathon, automating the process and reducing resource consumption by at **least 70%; received organization-wide recognition and bounty.**

SKILLS

- **Languages:** Python, Golang, C++, Java, Scala, SQL, JavaScript/TypeScript, Rust, R
- **Cloud & Infrastructure:** AWS, GCP, Kubernetes, Docker, Kafka, Linux, CI/CD, Jenkins, Databricks
- **Data & ML Technologies:** Spark, Airflow, PyTorch, TensorFlow, MongoDB, Redis, MySQL, PostgreSQL, DynamoDB
- **Extra-curricular/Awards:** Head Chess Club IITI, Exceptional Work Award for HMS Upgrade, Inter-IIT Sports Meet.